Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/29/2018
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Biofilm Formation, Growth, and Gene Expression on Different Materials and Environmental Conditions in Microgravity		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	01/03/2017	**End Date:**	01/02/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Klaus, David Ph.D. (University of Colorado, Boulder) Moeller, Ralf Ph.D. (Deutsches Zentrum Fuer Luft- Und Raumfahrt E.V.) Muecklich, Frank Ph.D. (Saarland University) Stodieck, Louis Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	80NSSC17K0036 ; NNX17AC21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Biofilm formation and its concomitant risk of causing equipment malfunction and human illnesses is a problem that needs to be addressed to enable safe long-term human space exploration. With that objective, this proposal aims to characterize biofilm growth during one International Space Station (ISS) increment in terms of mass, thickness, morphology, and associated gene expression using different spaceflight-relevant microbial species and material substrata. This project also aims to elucidate the biomechanical and transcriptomic mechanisms involved in the formation of the “column-and-canopy” biofilm architecture observed in space. The experiment is designed to assess the expression of genes associated with conferring microorganisms with resistance to oxidative stress, acidity, and antimicrobials in biofilms formed in space, and to determine physical mechanisms of material/microorganisms interaction in biofilms. The experiment will be conducted using BioServe's BioCells housing substrata coupons exposed to different microorganisms and media. Post-flight analyses of fixed samples include confocal laser scanning microscopy and RNA-sequencing. The significance of this project stems from gaining valuable new knowledge that may lead to improved methods and/or material properties for controlling biofilm formation during long duration space missions. The selection of bacterial species, growth media, and substrata material make the results applicable to a) the safety and maintenance of crewed spacecraft, b) planetary protection, c) mitigation of biofilm-associated illnesses on the crew, and d) biofilm-associated illnesses on Earth, thus making it relevant not only to this Research Announcement, but also to NASA’s Human Exploration and Operations Directorate, Space Biology Program, and Human Research Roadmap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biofilms are associated with multiple illnesses, including endocarditis (bacterial infection of cardiac tissue), cystitis (a urinary tract infection), and otitis media (an inflammatory disease of the middle ear), as well as several other hospital-acquired infections. Understanding how these biofilms form and their relationship with resistance to antibiotics can help us better prepare novel solutions to these problems on Earth.
Task Progress:	Progress is reported for the bacterial and fungal components of this project, separately: BACTERIAL a) We down-selected the material-media combination to three media: 3% Tryptic Soy Broth (TSB), iodine solution, and mAUM. Materials still being considered for flight are: Stainless Steel, Silicone, Aluminum, Cellulose Membrane, Massachusetts Institute of Technology’s (MIT’s) Lubricant Impregnated Surface (LIS), and Titanium alloy b) MIT's lubricant impregnated surface (LIS) is now being considered for flight. We were unable to form bacterial biofilms on it, so it is a candidate for further interrogation c) Results from our cold-stowage tests indicate that bacteria can be stowed at 4C for up to 21 days between preparation at launch site and experiment activation on orbit. Additionally, media by itself is viable for at least six months d) Fluorescence microscopy protocols and data acquisition protocols have been developed, including an in-house MatLab code that yields (i) biomass, (ii) mean thickness, (iii) substratum coverage, and (iv) roughness e) Motility studies indicate PA01 is a non-motile strain f) Quantitative tests did not confirm prior biofilm characterization results, indicating that changes to the design variables are still needed, informed by further tests FUNGAL g) Potato Dextrose Broth (PDB) was down-selected for further studies h) Cold-stowage test inform that inoculum in PDB cannot be frozen, as it precipitates during thawing. PDB can be stowed at 4C but for no longer than six weeks, before it starts impacting fungal growth. Iodine solution can be safely stowed at 4C for up to the tested 13-month duration i) Quantitative analyses did not corroborate prior biofiolm formation tests, and it was decided to grow the biofilm in air instead of submerged in liquid. This (i) enables biofilm formation, (ii) makes our data comparable to Earth protocols, and (iii) best replicates when molds have been observed on Salyut 6, 7, Mir, and ISS j) Two dry inoculation protocols were tested - (i) droping a 10 uL drop of inoculum over the coupon and letting dry, and (ii) a stamping method where 5g are placed over the back of a coupon sitting over a 7-10 day-old colony of fungi on agar. The latter approach yielded the best biofilms k) The previous test was replicated with 100% relative humidity, further improving the robustness of the biofilms l) BioServe's flight hardware P-HAB was tested, allowing us to conclude it is appropriate to keep the humidity and let fungi biofilm grow m) RNA quantity and quality was assessed, concluding that RNAProtect Bacteria was not yielding good results, but RNALater produced enough RNA and enough RNA integrity number (RIN) for full RNASequencing n) We improved on the dry stamping protocol, further simplifying it and making our samples more relevant to the conditions surfaces experience on spacecraft.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Abstracts for Journals and Proceedings	Schauer R, Ganguly S, Nisar Z, Cortesão M, Moeller R, Luo J, Muecklich F, McBride SA, Varanasi KK, Sato K, Gorti S, Klaus D, Stodieck L, Zea L. "Tests in Preparation for a Spaceflight Fungal and Bacterial Biofilm Experiment." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018

Abstracts for Journals and Proceedings	Zea L. "Preparando un experimento de biopelículas microbianas a ser llevado a cabo en la Estación Espacial Internacional." CONVERCIENCIA, Guatemala City, Guatemala, July 23-27, 2018. CONVERCIENCIA, Guatemala City, Guatemala, July 23-27, 2018. , Jul-2018
Abstracts for Journals and Proceedings	Cortesao M, Luo J, Mueller D, Nisar Z, Rubin P, Mucklich F, Hemmersbach R, Hellweg C, Moeller R, Zea L, Moeller R. "Biofilm in Space (BFS): Designing a Spaceflight Experiment." Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. , Jul-2018
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Abstracts for Journals and Proceedings	Zea L. "Thinking Outside the Globe -- Using Microgravity to Find Novel Solutions to Medical Problems on Earth." Aerospace Engineering Sciences Faculty Candidate Seminar, University of Colorado, Boulder, April 30, 2018. Aerospace Engineering Sciences Faculty Candidate Seminar, University of Colorado, Boulder, April 30, 2018. , Apr-2018
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